Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/20/2013
PI Name:	Thiessen, David Brian Ph.D.		
Project Title:	Phase Separation and Phase Change Using Helically-Supported Capillary Channels		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/15/2014
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/15/2014 per NSSC information (Ed., 4/5/2013) NOTE: New end date is 5/31/2013 per NSSC (Ed., 4/18/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marston, Philip (Washington State University)		
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Task Description:	Wicking structures with low resistance to axial flow are currently used in heat pipes and may have other applications in microgravity associated with phase separation and phase change technologies. The proposed work involves a fundamental study of a novel capillary flow channel consisting of a helical-wire support that is potentially useful in technologies and benefits from having a symmetry that simplifies the analysis of stability and flow dynamics. The channel geometry makes it particularly easy to measure the capillary dynamics on a global scale using high-speed imaging and image analysis. It is proposed to study the capture of droplets from a flowing air stream by a liquid-filled channel and the capture of bubbles from a flowing liquid stream by a gas-filled channel. Phenomena to be understood include the drop or bubble coalescence event and the subsequent response of the channel as capillary waves propagate from the impact zone dissipating energy. A model for capillary wave dynamics and for dynamic flows in these channels will be developed. Models for drop and bubble capture will also be developed that will be coupled to the channel dynamics model. Studies of phase change at an extended meniscus supported by the helical wire are also proposed. In all cases the basic component to be studied is the capillary channel supported by a helical wire and connected to a constant-pressure reservoir. The behavior of multichannel arrays for phase separation will also be investigated. Parabolic flight experiments are planned and ISS facilities are identified for possible future experiments. The understanding gained from this project is expected to be applicable to other channels that are based on the same fundamental physics. The experimentally validated physics-based models developed from this work might help guide the design of heat pipes and of novel phase-separation devices.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The experimentally validated physics-based models developed from this work might help guide the design of heat pipes and of novel phase-separation devices. The channels being developed have potential as variable-pitch superhydrophobic channels with small resistance to flow.
Task Progress:	An important goal of the project is to establish the performance limits of capillary channels that use helical support structures so that decisions can be made as to the suitability of such structures for applications in space such as cryogen control in tanks or phase separation in Rankine-cycle power systems. Significant progress was made toward this goal in both experiment and modeling. On the experimental front we have shown that flow rates close to theoretical limits can be achieved in horizontal springs up to 0.25-inch diameter under 1g conditions. We have also made progress on phase-separation studies. Very interesting results were obtained for the efficiency of drop capture by a liquid-filled spring as a function of impact speed. We have also for the first time achieved very efficient bubble capture by a submerged air-filled spring coated with a superhydrophobic aerogel. On the modeling front we have validated a finite element model of fully-developed flow in a periodically-supported channel that uses periodic boundary conditions by comparing results to several independent semi-analytical models from the literature for closely-related geometries. Experimental Progress A test rig for measuring the steady flow properties of water in large aspect ratio spring channels (for 0.125-inch and 0.25-inch diameter springs) was modified to allow for higher flow rates and to allow for accurate mean channel pressure and pressure drop measurements. The flow rig consists of a diaphragm pump that pumps water from a low-pressure reservoir to a high-pressure reservoir. The reservoirs in turn are connected to the two ends of the channel through flow-development sections that allow for the flow to become fully developed prior to entering the channel. Pressure taps are positioned within 1.25 mm of each end of the channel for measuring pressure drop. A sensitive diaphragm pressure transducer (referenced to ambient) is used for measuring the pressure at each end of the channel independently during a steady-state flow experiment. Thermocouples in the reservoirs are used to log water temperature during the experiment since viscosity is strongly temperature dependent and will influence measurements. Calibrated rotameters are used to measure flow rate. Springs as large as 0.25-inch diameter with a length of 7 inches have been found capable of supporting flow rates as high as 600 mL/min without leaking when the spring is coated with a superhydrophobic aerogel. Hydrostatic tests on each channel are carried out prior to a flow test. The hydrostatic tests involve photographing the entire length of the channel at each of up to 10 pressures and then performing image analysis to measure the diameter profile of the channel as a function of pressure. The channel is then photographed with different steady flow rates and the diameter profile is measured and inverted to find the pressure profile in the channel. These profiles can give information on flow-development effects within the channel. Measurements of pressure drop versus flow rate show that the channel is behaving as a pipe with an effective slip boundary condition on the wall. At higher flow rates there is evidence of flow-development effects due to a change in the wall boundary condition as the flow leaves the solid-walled-pipe entrance section and enters the spring channel. The magnitude of the effective slip length for the channel is in agreement with predictions of our numerical model. Two undergraduate students worked on experimental projects this year. Connor Talbott performed a study of the fraction of water drops absorbed by a liquid-filled channel as a function of impact speed and channel pressure. He discovered that the efficiency of drop capture is invariant with impact speed up to a point and then drops off in a non-monotonic manner. Ian Van Houten coated a 0.125-inch diameter spring with superhydrophobic aerogel, immersed it in a tank of water, primed it by filling with air, vented it to the atmosphere, and showed that ~3 mm diameter bubbles released beneath the channel were nearly 100% absorbed up to very high bubble frequencies (several per second). He took high-speed videos of the bubble absorption event and found that in areas where the coating was not perfect, the bubble would impact the channel and bounce downward several millimeters before contacting the channel again and being absorbed. Modeling Progress Progress this year on modeling of steady flow in a spring channel has concentrated on model validation and automation of computations to allow for studying a large portion of parameter space. Semi-analytical models from the literature were coded in Matlab to produce results for comparison to our finite element model. One of the literature models (Lauga and Stone 2003) was for the case of a smooth tube with alternating slip and no-slip boundaries. The other literature model (Ng and Wang 2010) was for an axisymmetric geometry with stepped roughness and alternating slip and no-slip boundary conditions. Good agreement was obtained between our numerical model and the literature models for the respective geometries considered. This gives us confidence that the model works for the wire-ring geometry used to simulate flow in helically-supported capillary channels. Dissemination and Impact of Results

	A paper resulting from this project on capillary wave scattering by a ring barrier on a liquid cylinder that was submitted in the prior reporting period was revised and has now appeared in the Journal of Fluid Mechanics. A second paper on hydrostatic properties of liquid-filled springs supported primarily by a prior NASA grant has been completed and submitted to Physics of Fluids. A set of videos of phenomena relating to this project were displayed at the APS DFD conference in November 2012 and have been disseminated to the public through the eprint archive arXiv.org (http://arxiv.org/). The idea of using springs as either wetting-control microstructures or as larger scale liquid acquisition devices in cryogen tanks is part of a current proposal to NASA (STRO-ESI) with other WSU investigators and collaborators from NASA and a private aerospace firm. Students Impacted One physics major (Connor Talbott) completed his senior thesis on a project involving drop capture by a liquid-filled spring. Another physics major (Shen Wei) completed an independent-study project on finite element modeling of flow in twisted ducts. A chemical engineering student (Ian Van Houten) set up an experiment to capture bubbles by an air-filled spring and obtained high-speed videos of the process. A physics PhD student (Maverick Terrazas) received a NASA Space Grant fellowship to use the drop tower at Portland State University to study the design and performance of manifolds used to prime channels for phase-separation technologies. A chemical engineering PhD student (Baba Abdul) is currently finishing a PhD thesis on the steady flow experiments and modeling. References: E. Lauga and H. A. Stone, "Effective slip in pressure-driven stokes flow," J. Fluid Mech., Vol. 489, 55-77 (2003). C. -O. Ng and C. Y. Wang, "Stokes flow through periodically grooved tube," J. Fluids Eng., Vol. 132, 101204 (2010).
Bibliography Type:	Description: (Last Updated: 09/14/2018)
Abstracts for Journals and Proceedings	Bernate J, Thiessen DB. "Stability of capillary surfaces supported by a helical wire." Video submission to the APS DFD Gallery of the fluid motion 2012. Gallery of Fluid Motion, 65th Annual Meeting American Physical Society Division of Fluid Dynamics, San Diego, CA, November 18-20, 2012. Video submission to the APS DFD Gallery of the fluid motion 2012: http://arxiv.org/abs/1210.4100 (May 2013) ; will be available at Physics of Fluids website in the near future. , May-2013
Abstracts for Journals and Proceedings	Terrazas M, Thiessen DB. "Phase Separation by a Helically - Supported Capillary Channel Array: Parabolic Flight Experiments." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 28 - December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 28 - December 2, 2012. Program and abstracts, p. 105. https://asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Dec-2012
Articles in Peer-reviewed Journals	Zhang L, Thiessen DB. "Capillary-wave scattering from an infinitesimal barrier and dissipation at dynamic contact lines." Journal of Fluid Mechanics. 2013 Mar;719:295-313. http://dx.doi.org/10.1017/jfm.2013.5 , Mar-2013
Articles in Peer-reviewed Journals	Bernate J, Thiessen DB. "The stability of capillary surfaces on helical wire supports." Physics of Fluids. In press as of May 2013. , May-2013